

UNIT-I [RANDOM VARIABLES]

INTRODUCTION

When we toss a coin we get two outputs, head or tail. Instead of dealing the outcomes descriptively as head or tail we can assign numerical value say 1 to head and 0 to tail. This rule is from the original sample space to a numerical sample space to a certain constraints called Random variable.

Definition of Random Variable:-

A real variable X whose value is determined by the outcome of a random experiment is called a random variable. Random variables are of two types, namely

- (i) Discrete Random variable
- (ii) Continuous Random variable

Discrete Random Variable

A random variable X is said to be discrete if the set of values of X is finite or countable.

eg: (1) A coin is tossed 3 times

(2) The number of telephone calls are received from the telephone operator.

Continuous Random Variable

If a random variable takes on all values within a certain interval then the random variable is called continuous random variable. example:

$R.V. X =$ duration of telephonic conversation

Here X have all values (integral as well as fractional)

Probability mass function

Let X be a discrete random variable taking the values $x_1, x_2, \dots, x_n$ such that

$$P(X=x_i) = P_i$$

The P_i is the probability function (or) probability mass function (P.M.F)

P_i ($i=1, 2, \dots$) satisfies the following conditions

$$(i) P_i \geq 0 \text{ for all } i$$

$$(ii) \sum_{i=1}^{\infty} P_i = 1$$

$$(iii) 0 \leq P_i \leq 1$$

① If the Random variable (RV) X has the following probability

$X=x$	0	1	2	3	4	5	6	7
$P(x)$	0	K	$2K$	$2K$	$3K$	K^2	$2K^2$	$7K^2+K$
	0	$\frac{K}{10}$	$\frac{2K}{10}$	$\frac{2K}{10}$	$\frac{3K}{10}$	$\frac{K^2}{100}$	$\frac{2K^2}{100}$	$\frac{7K^2+K}{100+10}$

(i) Find the value of K .

(ii) $P(X < 6)$, $P(X \geq 6)$ and $P(0 < X < 5)$

(iii) Find the distribution function of X

(iv) Minimum value of a such that

$$P(X \leq a) > \frac{1}{2}$$

Solution:

$$(i) \text{ w.k.t } \sum_{i=1}^{\infty} P_i = 1$$

$$0 + K + 2K + 2K + 3K + K^2 + 2K^2 + 7K^2 + K = 1$$

$$10K^2 + 9K = 1$$

$$10K^2 + 9K - 1 = 0$$

$$10K^2 + 10K - K - 1 = 0$$

$$10K(K+1) - 1(K+1) = 0$$

$$(10K-1)(K+1) = 0$$

$$K = \frac{1}{10}$$

$$(or) K = -1$$

$\therefore K = -1$ is not possible

$$(ii) P(X \geq 6)$$

$$P(X \geq 6) = P(X=6) + P(X=7)$$

$$= \frac{2}{100} + \frac{1}{100} + \frac{1}{10}$$

$$= \frac{2}{100} + \frac{1}{100} + \frac{10}{100} = \frac{13}{100} = 0.13$$

$$P(X \geq 6) = 0.19$$

$$P(X < 6) = 1 - P(X \geq 6)$$

$$= 1 - 0.19$$

$$P(X < 6) = 0.81$$

$$P(0 < X < 5) = P(X=1) + P(X=2) + P(X=3) + P(X=4)$$

$$= \frac{1}{10} + \frac{2}{10} + \frac{2}{10} + \frac{3}{10}$$

$$P(0 < X < 5) = \frac{8}{10} = 0.8$$

(iii) Distribution function

$X=x$	0	1	2	3	4	5	6	7
$P(x)$	0	0.1	0.2	0.2	0.3	0.01	0.02	0.17
$F(x)$	0	0.1	0.3	0.5	0.8	0.81	0.83	1.00

Minimum value of a such that $P(X \leq a) > \frac{1}{2}$

$$a = 4$$

$\therefore a = 4$ in $P(X \leq a)$. Find value which is greater than $\frac{1}{2}$

$$(v) P(1.5 < X < 4.5 | X > 2) \\ = \frac{P(1.5 < X < 4.5 \cap X > 2)}{P(X > 2)}$$

$$= \frac{P(X=3) + P(X=4)}{P(X=3) + P(X=4) + P(X=5) + P(X=6) + P(X=7)}$$

$$= \frac{0.2 + 0.3}{0.2 + 0.3 + 0.01 + 0.02 + 0.17}$$

$$= \frac{0.5}{0.7} = 0.71$$

- ② If the R.V. X takes the values 1, 2, 3 and 4 such that $2P(X=1) = 3P(X=2) = P(X=3) = 5P(X=4)$. Find the probability distribution table and cdf of X .

Solution:

$$\text{Let } 2P(X=1) = 3P(X=2) = P(X=3) = 5P(X=4) = k \text{ (say)}$$

$$\begin{array}{|l} 2P(X=1) = k \\ P(X=1) = k/2 \end{array} \quad \begin{array}{|l} 3P(X=2) = k \\ P(X=2) = k/3 \end{array} \quad \begin{array}{|l} P(X=3) = k \\ 5P(X=4) = k \\ P(X=4) = k/5 \end{array}$$

X	1	2	3	4
$P(x)$	$k/2$	$k/3$	k	$k/5$

$$\text{w.k.t } \sum_{i=1}^{\infty} P_i = 1$$

$$\frac{k}{2} + \frac{k}{3} + k + \frac{k}{5} = 1$$

$$\frac{15k + 10k + 30k + 6k}{30} = 1$$

$$\frac{61k}{30} = 1$$

$$k = \frac{30}{61} \Rightarrow \boxed{k = 0.49}$$

x	1	2	3	4
$P(x)$	0.245	0.163	0.49	0.098
$F(x)$	0.245	0.408	0.898	0.996

- ③ The probability function of an infinite discrete distribution is given by $P(X=j) = \frac{1}{2^j}$ ($j=1, 2, \dots, \infty$). Verify that the total probability is 1 and find the mean and variance of the distribution. Find also $P(X \text{ is even})$, $P(X \geq 5)$, and $P(X \text{ is divisible by } 3)$.

Solution:

$$\text{Given } P(X=j) = P_j$$

TOTAL PROBABILITY

$$\sum_{j=1}^{\infty} P_j = \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \dots \infty$$

$$= \frac{1/2}{1 - 1/2}$$

$$= 1$$

Geometric Series
 $a + ar + ar^2 + \dots$
 $\therefore = \frac{a}{1-r}$

MEAN OF X

$$E(X) = \sum_{j=1}^{\infty} j P_j$$

$$= \frac{1}{2} + 2 \cdot \frac{1}{2^2} + 3 \cdot \frac{1}{2^3} + 4 \cdot \frac{1}{2^4} + \dots \infty$$

$$= \frac{1}{2} \left[1 + 2 \cdot \frac{1}{2} + 3 \cdot \frac{1}{2^2} + 4 \cdot \frac{1}{2^3} + \dots \infty \right]$$

$$= \frac{1}{2} \left[1 - \frac{1}{2} \right]^{-2}$$

$$= \frac{\frac{1}{2}}{\left(\frac{1}{2}\right)^2} = 2$$

VARIANCE OF X

$$\text{VAR}(X) = E(X^2) - (E(X))^2$$

$$E(X^2) = \sum_{j=1}^{\infty} [j^2 P_j]$$

$$= \sum_{j=1}^{\infty} [j(j+1) - j] P_j$$

$$= \sum_{j=1}^{\infty} j(j+1) \frac{1}{2^j} - \sum_{j=1}^{\infty} j \frac{1}{2^j}$$

$$= 2 \cdot \frac{1}{2} + 2 \cdot 3 \cdot \frac{1}{2^2} + 3 \cdot 4 \cdot \frac{1}{2^3} + \dots$$

$$- \left(\frac{1}{2} + 2 \cdot \frac{1}{2^2} + 3 \cdot \frac{1}{2^3} + \dots \right)$$

$$= \frac{1}{2} \cdot 2 \left(1 + 3 \cdot \frac{1}{2} + 3 \cdot 2 \cdot \frac{1}{2^2} + \dots \right)$$

$$- \frac{1}{2} \left(1 + 2 \cdot \frac{1}{2} + 3 \cdot \frac{1}{2^2} + \dots \right)$$

$$= (1 - \frac{1}{2})^{-3} - \frac{1}{2} (1 - \frac{1}{2})^{-2}$$

$$= \left(\frac{1}{2}\right)^{-3} - \frac{1}{2} \left(\frac{1}{2}\right)^{-2}$$

$$= 2^3 - \frac{2^2}{2} = 2^3 - 2 = 8 - 2 = 6$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

$$= 6 - (2)^2$$

$$= 6 - 4$$

$$= 2$$

$$P(X \text{ is even}) = P(X=2 \text{ or } X=4 \text{ or } X=6 \text{ or etc})$$

$$= P(X=2) + P(X=4) + \dots$$

$$= \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^4 + \dots$$

$$= \frac{\frac{1}{4}}{1 - \frac{1}{4}} = \frac{1}{3}$$

$$P(X \text{ is even}) = \frac{1}{3}$$

$$P(X \geq 5) = P(X=5) + P(X=6) + \dots$$

$$= \left(\frac{1}{2}\right)^5 + \left(\frac{1}{2}\right)^6 + \dots$$

$$= \frac{\left(\frac{1}{2}\right)^5}{1 - \frac{1}{2}} = \frac{1}{16}$$

$$P(X \geq 5) = \frac{1}{16}$$

$$P(X \text{ is divisible by } 3)$$

$$= P(X=3 \text{ or } X=6 \text{ or } X=9 \text{ etc})$$

$$= P(X=3) + P(X=6) + \dots$$

$$= \left(\frac{1}{2}\right)^3 + \left(\frac{1}{2}\right)^6 + \dots$$

$$= \frac{\frac{1}{8}}{1 - \frac{1}{8}} = \frac{1}{7}$$

$$P(X \text{ is divisible by } 3) = \frac{1}{7}$$